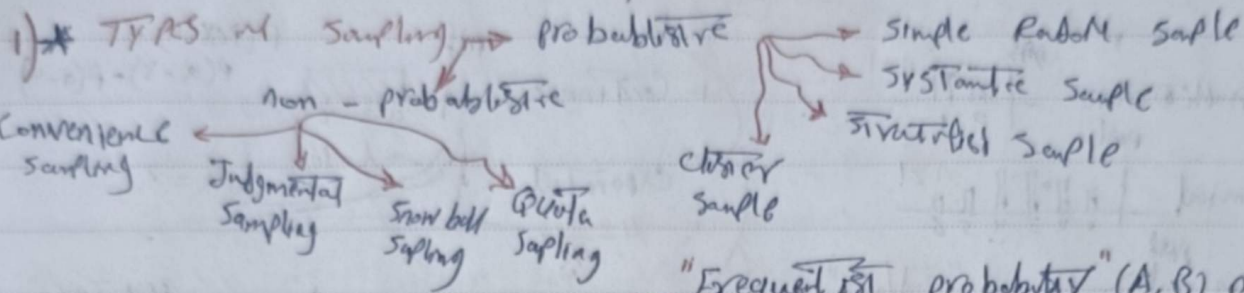


* Types of Sampling - Probability - Variable - Data - Statistics



2) * Types of Probability

"Marginal probability"

(A, B) are independent

$$P(A) = \sum_b P(A \cap B = b)$$

"Conditional Prob"

$$P(X|Y) = \frac{P(A \cap B)}{P(B)}$$

"Frequentist probability" (A, B) not in the same time $P(S) = 1$ $P(A) \neq 0$

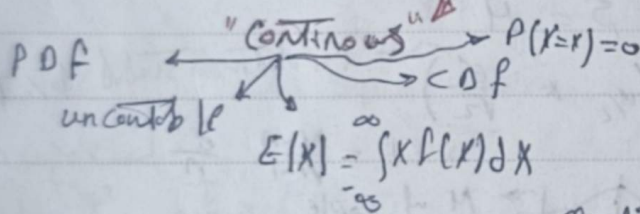
$$P(A \cup B) = P(A) + P(B)$$

"Joint Probability" (A, B) can be in the same time

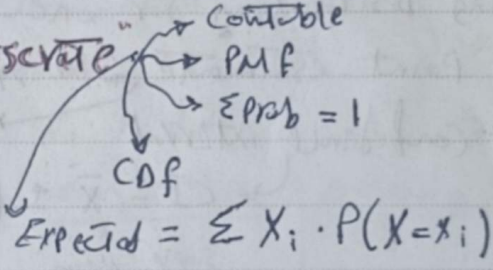
$$P(A \cap B) = P(A) \cdot P(B)$$

Can be for multiple events

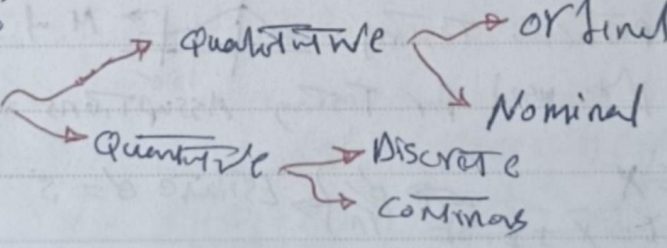
3) * Types of Variables



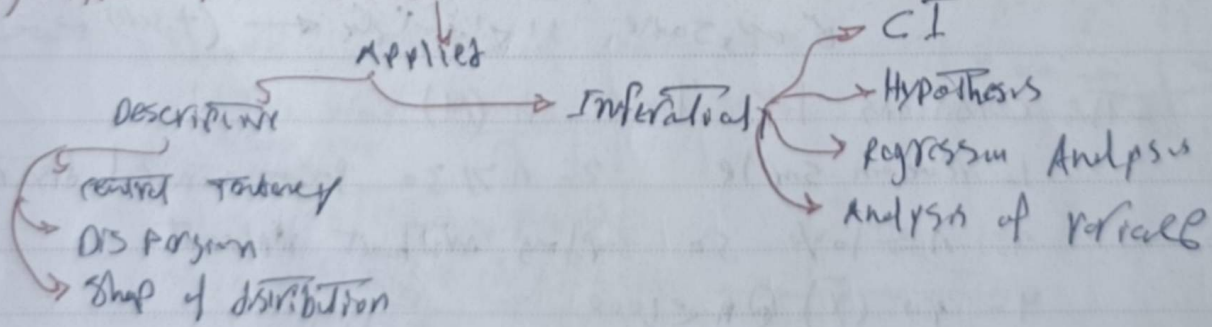
"Discrete"



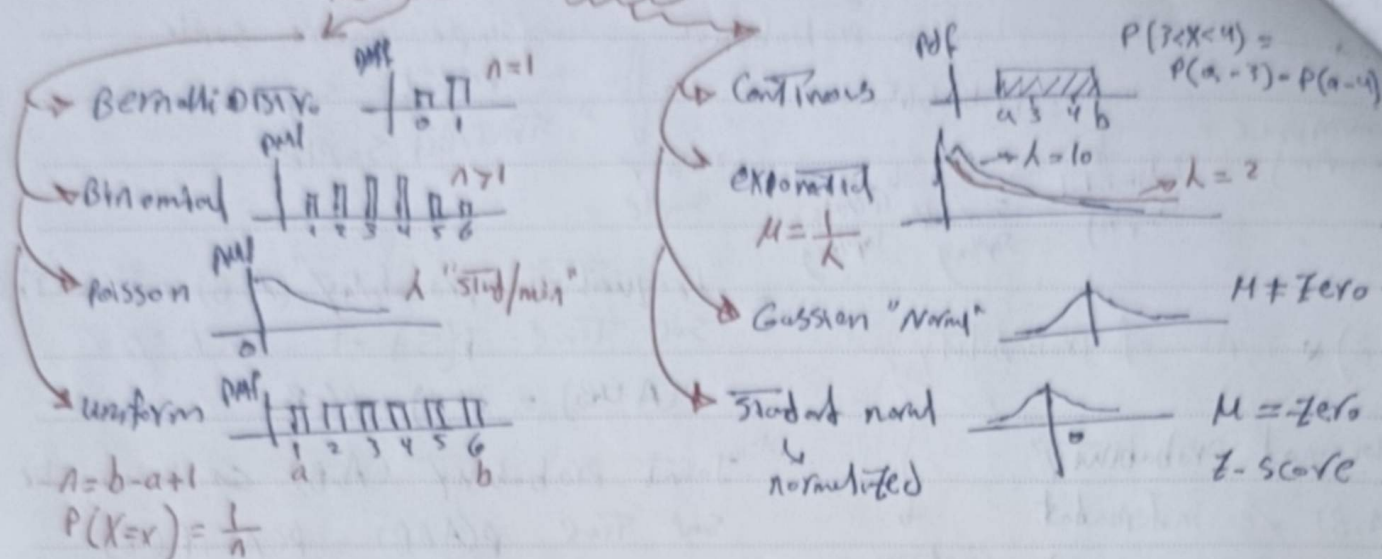
4) * Types of Data



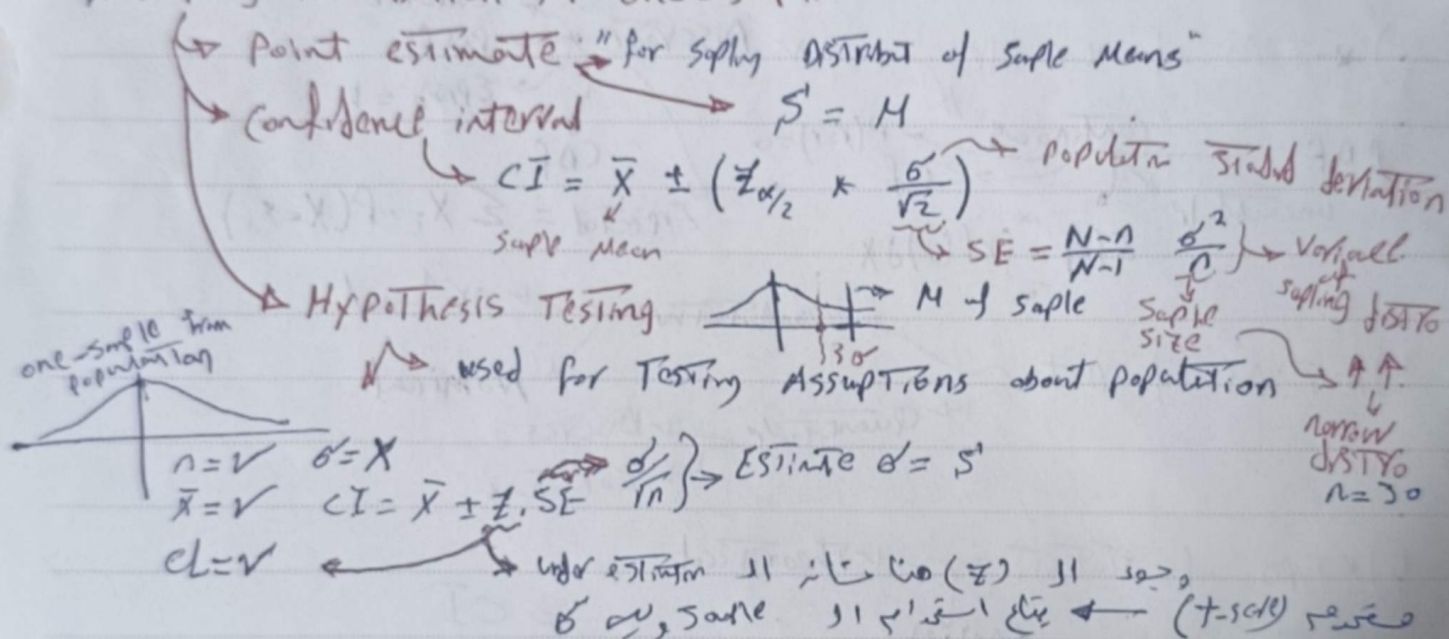
5) * Types of Statistics



* Types of Discrete & Continuous Distribution



* Sampling Distribution for "one-sample"



The conditions for inference on (μ)

1. Random sample
2. $n \geq 30$ for non-normal distro
3. $n \leq 10\%$ so sampling without replacement
4. $var(\bar{X}) @ n < 1000 = \frac{\sigma^2}{n-dof}$

SENA

Type I, II error

true state of nature	Decision Made	
	Ho	Ha
Ho	correct decision $1-\alpha$	Type-1 error α
Ha	Type-2 error β	correct decision $1-\beta$

Decreasing (α) will increase (β)
($1-\beta$) $\uparrow$ α $\downarrow$ β $\downarrow$ $1-\beta$ $\uparrow$

* Sampling distribution for MULTIVARIATE

Comparing proportion between 2 independent group
Ho: $p_{diff} = 0$ Z-TEST $\rightarrow$ p-value

Gender	Voting
M	F
$p_{diff} = p_M - p_F$	

Comparing Mean between categorical & continuous
Ho: $\bar{M}_{diff} = 0$ t-TEST $\rightarrow$ p-value

Gender	Survival
M	F
$\bar{X} = \bar{M}_{diff} = \bar{X}_M - \bar{X}_F$	

dot = $n_1 + n_2 - 2$ no of groups

one-sample paired t-test "Before, After" diff

T-TEST $\rightarrow$ p-value

dot = $n - 1$ Ho: no diff

Chi Square Test

3 proportions p_1, p_2, p_3

Chi-TEST $\rightarrow$ p-value

Passenger class	Survival
1	0
2	1
3	0
1	0

2 Categorical Variables

goodness of fit

dot = $(\# \text{Categorical} - 1)$

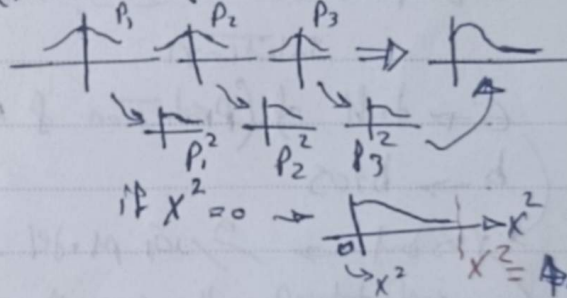
For independence $\rightarrow$ Survival classes

dot = $(\# \text{rows} - 1)(\# \text{columns} - 1)$

tdis

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Ho: $p_1 = p_2 = p_3$



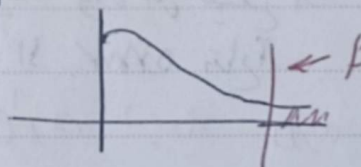
ANOVA $\rightarrow$ extension of compare mean

Two-Way	Gender	Class	Age
one-way	M	1	2
Multi	F	2	3

F-TEST $\rightarrow$ p-value

$$F = \frac{SSB}{SSW} \rightarrow \left(\frac{s_1}{s_2} \right)^2 = \text{chi}$$

$$\left(\frac{s_2}{s_2} \right)^2 = \text{chi}$$



Correlation Test

S Test $\rightarrow$ P-value
T Test

$H_0: \rho = 0$

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{(n \sum X^2 - (\sum X)^2)(n \sum Y^2 - (\sum Y)^2)}}$$

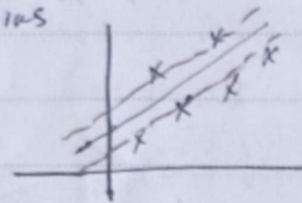
$\rho_{X,Y} = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$
Covariance

Cov $\rightarrow$ give indication of the correlation (-1, 1)
corr $\rightarrow$ give the strength
Regression $\rightarrow$ Amount of change
corr signified Test

Regression Analysis $\hat{Y} = f(X)$ from sample

SLR
 $\hat{Y} = aX + b$

$b \rightarrow$ bias



$a, b \rightarrow$ point estimate
CI

a, b are sample mean
sample mean
population

estimation by error term
 $Y = aX + b + \epsilon$
sample mean $(\bar{Y})$
N of point estimate
solution

$\hat{Y}$ diff of (prediction & Actual)

$b \rightarrow$ bias

residual $\rightarrow$ Model error
overfitting
training error

a, b are

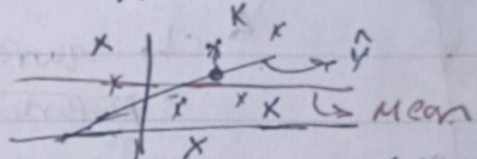
Multi linear regression $\hat{Y} = f(X)$

$\hat{Y} = a_0 + a_1 X_1 + a_2 X_2 + \dots + a_n X_n$
 $\hat{Y} = X^T \cdot b + \epsilon$

using OLS

$b_{OLS} = (X^T X)^{-1} X^T Y$

optimization method
Algebraic



$(R^2) \leftarrow$ Coeff of determination
error

MSE, MAE, RMSE
error

Model error

Mean

Mean

$SST = \sum (Y - \bar{Y})^2$ Model error

$SSR = \sum (\hat{Y} - \bar{Y})^2$ Total explained

$SSR_s = \sum (\hat{Y} - Y)^2$ unexplained

$R^2 = 1 - \frac{SSR_s}{SST}$

$SSR_s = SST - SSR$
Mean

* يجب ان R^2 انما يتبع خاصية واحدة لو كانت R^2 Adjusted R^2 $0 \leq R^2 \leq 1$ و $0 \leq R^2 \leq 1$

$$R_{AD}^2 = 1 - \frac{(1 - R^2)(N-1)}{N-p-1}$$

→ number of features

والتي انما يتم نقل R_{AD}^2 نقل اما في حالة التفسير والى ان يتفقا كمتة واحدة ولا في الابط (R^2) مزيد، والى انما الابط نقل، و R_{AD}^2 ← R^2

Significance Test of

Coeff

$b_0, b_1, \dots, b_n$

Significance test

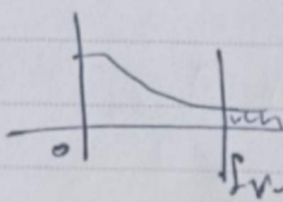
of parameter

T-Test and

error follow normal distr

$H_0: b = 0$

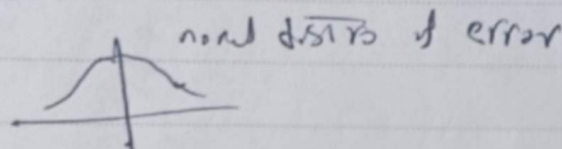
$$t = \frac{b - 0}{SE}$$



$H_0: R^2 = 0 \rightarrow \text{Model} = \text{Mean}$

$$F_{stat} = \frac{(R^2 - 0) \cdot (n - k - 1)}{(1 - R^2) \cdot k}$$

$p < \alpha \rightarrow \text{reject } H_0$



normal distr of error

Linear Algebra

التاريخ:

الموضوع:

* Vector & Matrix operations:-

1) Types of vectors: Row vector & Column vector

$$V_{\text{row}} \times M \quad M \times V_{\text{col}}$$

2) Types of Matrix: Identity & Diagonal & scalar & symmetric

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 & 0 \\ 0 & 4 & 0 \end{bmatrix} \quad \begin{bmatrix} 5 & 5 & 0 \\ 0 & 5 & 0 \end{bmatrix} \quad A = A^T$$

Ref & RREF & upper Triangle & lower Triangle & singular Matrix

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1-3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad \begin{bmatrix} 8 & 2 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 4 \end{bmatrix} \quad \begin{bmatrix} 3 & 0 & 0 \\ 1 & 5 & 4 \\ 4 & 2 & 1 \end{bmatrix} \quad \det(A) = 24$$

Rank < Dimension

3) vector operations

- Addition and subtraction - Dot product
- scalar Multiplication - cross product (3D)
- Vector Norms

Ex $\vec{v}_1 \cdot \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = (1 \times 2) + (1 \times 3) + (1 \times 4)$

Ex $\vec{v}_1 \times \vec{v}_2 = \begin{bmatrix} i & j & k \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix}$
 → give 3 component in 3D
 → $|A||B| \sin \theta$ → give vector $\perp (v_1, v_2)$

Ex L1 Norm - L2 Norm - L ∞ Norm - LP Norm

Manhattan - Euclidean - Max - Generalized

$$\|v\|_1 = \sum_{i=1}^n |v_i| \quad \|v\|_2 = \left(\sum_{i=1}^n v_i^2 \right)^{1/2} \quad \|v\|_\infty = \max(|v_1|, |v_2|, \dots, |v_n|) \quad \|v\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

$$= |3| + |-4| + |-2| = 9 \quad \sqrt{3^2 + (-4)^2 + (-2)^2} = \sqrt{29} \quad \max(|3|, |-4|, |-2|) = 4$$

4) Matrix operations

- Addition and subtraction - Transpose
- scalar Multiplication - Determinant
- Matrix Multiplication - inverse

Ex $A \cdot B = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$ $2 \times 3 \quad 3 \times 2 \quad 2 \times 3$
 $mxn \quad nxL \quad mxL$
 Ex $A^T = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}$ where $A = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$

Ex $A \cdot A^{-1} = I$

square

* Vector and Matrix Concepts /-

1) MSE and MAE

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n} \|\hat{y} - y\|_2^2$$

↓ sensitive To outliers - Ridge operator

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| = \frac{1}{n} \|\hat{y} - y\|_1$$

efficient with outliers - Lasso operator

2) Dot product Applications

Dimensionality Reduction using PCA
To project data in (n-D) To (2D)

SVD

Eigen decomp of Cov Matrix

Projection Proj_b a = $\frac{a \cdot b}{b \cdot b} b \rightarrow$ Dimensionality reduction

optimization of Gradient Descent

Normalization

Cos Similarity $\{a, b\} = \frac{a \cdot b}{\|a\| \|b\|} \rightarrow$ Text Analysis and recommendations systems where orientation is more important than magnitude.

Vector normalization for feature scaling $\vec{a} = \frac{a}{\|a\|}$

Changing basis or coordinates of Robotics & Signal and control

3) Vector (Space, Span, Basis) and Linear (sys eq, combination, independence and Transformation)

Vector Space

vector space $\rightarrow$ known as inner space

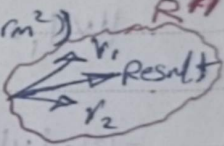
function space $\rightarrow f(x) = Ax + b$

(A) is a matrix & (b) is a vector

polynomial space $\rightarrow P(x) = a_0 + a_1x + \dots + a_nx^n$

Euclidean space (Norm²)

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



Linear Combination

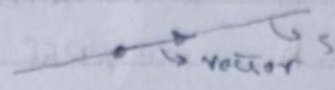
$L = c_1v_1 + c_2v_2 + c_3v_3 \rightarrow$ "L" is linear combination of (v_1, v_2, v_3)

$L \in \text{Span}(v_1, v_2, v_3)$ "3D"

$v_1, v_2, v_3 \rightarrow$ linearly independent vectors if every single vector add new dimensions in the Span

$v_1, v_2, v_3 \rightarrow$ linearly dependent $\rightarrow$ at least one vector can be expressed as linear combination of the others

if W not linear combination from $v_1, v_2 \rightarrow$ we can not write $W = av_1 + bv_2$
 $W \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \neq \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \rightarrow a, b \rightarrow$ Trivial solution

The span of vector is a line  span (R^1)

if v_3 not in span of $(v_1, v_2) \rightarrow v_3$ not in the plane "space" of $(v_1, v_2) (R^2)$

No vector in the basis can be expressed as a linear combination of the others $(v_3 \neq a_1v_1 + a_2v_2)$ or $(v_1 \neq a_2v_2 + a_3v_3)$

if 3 vector linearly independent $\rightarrow$ 3 vector not linear combination
 $C_1v_1 + C_2v_2 + C_3v_3 \neq 0$

If 3 vectors are dependency $\rightarrow$ Existing non-trivial solution

$$v_3 = -\frac{C_1}{C_3}v_1 - \frac{C_2}{C_3}v_2 \text{ (if } C_3 \neq 0)$$

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad v_2 = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix} \quad v_3 = \begin{bmatrix} 3 & 5 & 7 \end{bmatrix}$$

$$v_2 = 2v_1 + 0v_3 \quad \therefore v_2 \text{ depends of } v_1 \text{ only}$$

if (v_1, v_2) are indepent $\rightarrow (v_1, v_2)$ are Basis of R^2

Types of Basis [orthogonal - non orthogonal - orthonormal]

Preserve Linearity - origin

used for scaling - reflection -

vector affected $T_e \cdot v_e = u_e$ where $T \rightarrow$ new basis with respect to (e)

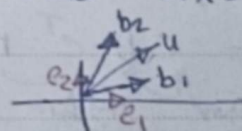
SYS of linear Transformation revisited if $AX=b \rightarrow X=A^{-1}b$

$$\text{if } T_e \cdot u_b = u_e \quad \therefore u_b = T_e^{-1} u_e$$

we use chng basis for Diagonalization & Dimensionality Reduction

$$\hat{u}_b = B_e^{-1} \cdot T \cdot B_e \cdot u_b$$

revisited Transformation respect e basis
 convert u_b to u_e



* Linear Algebra Application 1

1) Matrix inversion & Gaussian elimination

Inverse $\rightarrow Ax = b \rightarrow x = A^{-1} \cdot b$ case $A \rightarrow$ square matrix

Gaussian $\rightarrow Ax = b$ use RREF used if non-square or singular matrix

2) Matrix Rank & Determinant & RREF

- used All of them in checking if vectors are independent
- no of Rank = no of indep vector "no feature depend on other"
- Matrix Determinant $\rightarrow$ in 2D "Area" & 3D "Volume"

$$\text{Area}_f = \det \cdot \text{Area}_e$$

if $\det = 0 \rightarrow$ The Area in $\mathbb{R}^2$ collapsed in $\mathbb{R}^1$ on a line

cause 2 vectors are dependent $\rightarrow (T) \rightarrow$ singular

3) Orthogonalization

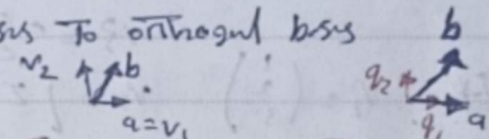
- Convert from non-orthogonal basis to orthogonal basis

- using Gram-Schmidt

$$\hat{v}_1 = \frac{a \cdot v_1}{v_1 \cdot v_1} \quad \hat{v} = \frac{v_1}{\|v_1\|} \quad \# \text{ } a_1 = v_1$$

$$q_2 = \frac{\hat{v}_2}{\|\hat{v}_2\|} \quad \hat{b} = \hat{v}_2 + \text{proj}_{v_1} b$$

$$q_3 = \frac{\hat{v}_3}{\|\hat{v}_3\|} \quad \hat{c} = \hat{v}_3 + \text{proj}_{v_1} c + \text{proj}_{v_2} c$$



$$E = [q_1 \ q_2 \ q_3] \quad \therefore \bar{r}_b = E \cdot T_E \cdot E^{-1} \cdot r_b$$

$$E^{-1} = E^T = B$$

4) eigen value & vectors (1 $\rightarrow$ value)

After Transformation the vector in the same space "scaled only"

Diagonalization $\rightarrow$

using eigen decomposition

$$r^1 = C D C^T r$$

return to original $\leftarrow$ Transform into eigen basis $\leftarrow$ from original to eigen

non orthogonal $\rightarrow$ orthogonal basis $\leftarrow$ id $\leftarrow$ C.T $\leftarrow$ eigen base $\leftarrow$ where

SENA

$$D^{1000} = \begin{pmatrix} \lambda_1^{1000} & 0 & 0 \\ 0 & \lambda_2^{1000} & 0 \\ 0 & 0 & \lambda_3^{1000} \end{pmatrix}$$

Dimensionality Reduction $\rightarrow$ PCA

using eigen decomposition for Cov

Cov matrix $\rightarrow$ symmetric matrix

$$\text{full orthogonal } Q^{-1} = Q^T \quad \lambda \gg 0$$

$$\text{corr} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sum (x_i - \mu_1)(y_i - \mu_2)}{n-1 \sigma_X \sigma_Y}$$

- 1) standard data
- 2) cov
- 3) eigens calculation
- 4) sorting
- 5) pick biggest values

First order

التاريخ:

Optimization

الموضوع

* Batch Gradient Descent $\theta_{t+1} = \theta_t - \alpha \nabla \theta_t$

* Stochastic Gradient Descent

* Mini Batch Gradient Descent

* Momentum-Based Gradient Descent $v_t = \beta v_{t-1} + \alpha \nabla \theta_t$ * Nesterov Accelerated Gradient $\theta_{t+1} = \theta_t - v_t$

$$\theta_{Temp} = \theta_t - \beta v_{t-1}$$

$$v_t = \beta v_{t-1} + \alpha \nabla \theta_{Temp}$$

$$\theta_{t+1} = \theta_{Temp} - \alpha \nabla \theta_{Temp}$$

* Adaptive Gradient Methods "helpful for dense & sparse features"

$$v_t = v_{t-1} + (\nabla \theta_t)^2 \quad \theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{v_t}} \nabla \theta_t$$

$$\text{Ada grad method } v_t = v_{t-1} + (\nabla \theta_t)^2 \quad \theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{v_t} + \epsilon} \nabla \theta_t$$

$$\text{RMS Prob Method } v_t = \beta v_{t-1} + (1-\beta)(\nabla \theta_t)^2 \quad \theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{v_t} + \epsilon} \nabla \theta_t$$

$$\text{Adam Method } m_t = \beta_1 m_{t-1} + (1-\beta_1)(\nabla \theta_t) \quad v_t = \beta_2 v_{t-1} + (1-\beta_2)(\nabla \theta_t)^2$$
$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{v_t} + \epsilon} \hat{m}_t \quad \hat{m}_t = \frac{m_t}{1-\beta_1^t} \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t}$$

→ Momentum of NAG → Can not handle sparse features

→ Ada & RMS prob → Can not handle vanishing

→ Adam → Can handle both

→ First Moment → For the mean of gradients → capture avg direction over time

→ Second Moment → For the variance of gradients → capture magnitude

$$\text{Taylor series } f(x_{k+1}) \approx f(x_k) + f'(x_k)(x_{k+1} - x_k)$$
$$\nabla f = \frac{-f(x_k)}{f'(x_k)} = x_{k+1} - x_k \Rightarrow \nabla f = -J(\theta) \quad \text{Newton}$$

$$J(\theta_{t+1}) = J(\theta_t) + J'(\theta_t)(\theta_{t+1} - \theta_t) \quad \text{gradient}$$
$$\nabla \theta_{t+1} = J(\theta_t) + \nabla(\theta_t) \cdot \Delta(\theta_t) \quad \text{must be negative of dot product (-)}$$

$$\nabla \theta_t \propto -J(\theta_t) \quad \text{gradient}$$

$$\theta_{t+1} = \theta_t - \nabla \theta_t$$

Opposite of gradient

second order

التاريخ:

الموضوع:

- * Newton's method $\rightarrow$ will approximate cost function as a curve $\rightarrow$ suitable for convex only
 - * single variable $\theta_{t+1} = \theta_t - \frac{J'(\theta_t)}{J''(\theta_t)}$
 - * Multi variables (n-dimension) $\theta_{t+1} = \theta_t - H^{-1}(\theta_t) \nabla f(\theta_t)$
- Can't find the local min, max, saddle computationally expensive "inverse"

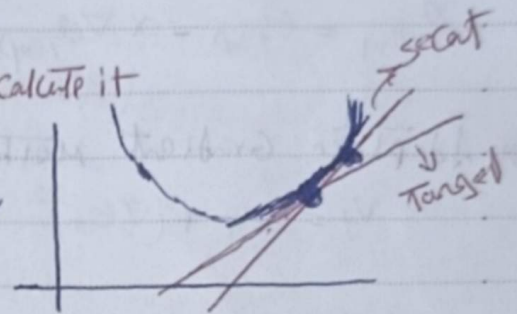
Enhanced $\left\{ \right.$ Newton's method

* Quasi-Newton Methods

We will approximate Hessians instead calculate it

Tangent $\rightarrow$ derivative

secant $\rightarrow$ Approximation of derivative



$$\begin{aligned} f(x_{t+1}) - f(x_t) &= f'(x_t)(x_{t+1} - x_t) \\ &\approx H_{t+1} s_t = y_t \\ &\rightarrow \nabla f(\theta_{t+1}) - \nabla f(\theta_t) \end{aligned} \quad \Rightarrow \text{secant condition}$$

Approximation $\theta_{t+1} - \theta_t$

Hessian Assumptions

$$g = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\theta_0 = \begin{bmatrix} 0.001 \\ 0.001 \\ 1 \\ 0.001 \end{bmatrix}$$

$$B_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\theta_{t+1} = \theta_t - \alpha B_t^{-1} \nabla f(\theta_t)$$

But here function not sensitive like before for α .

* BFGS $\rightarrow$ Algorithm used to put some constraints to solve

$$[B_t^{-1}]$$

Constraints $\parallel$ ليس له أي شيء